

Do Higher Temperatures Causally Affect Walmart Weekly Sales?

Ashar Hashmi

Table of contents

1. Introduction

Walmart wants to know whether rising temperatures might hurt or help their weekly sales. With temperatures going up over time, this question matters for planning and forecasting.

I have weekly data from 45 Walmart stores covering February 2010 through October 2012. For each store and week, I observe:

- Weekly sales
- Temperature
- Whether it was a holiday week
- Fuel price, CPI, and unemployment

Research Question: In these stores, do higher temperatures causally affect weekly sales? The main challenge is that both temperature and sales are seasonal. For instance, temperatures are higher in summer, and sales also change across the year because of holidays and shopping patterns. If we simply compare hot weeks to cold weeks, we might mistake seasonality for a temperature effect.

To handle this, I use a panel data approach with:

- Store fixed effects: They control for any permanent differences between stores (like size, location, customer base).
- Date fixed effects: They control for anything that happens in the same week for all stores, such as national holidays, seasonality, or macroeconomic shocks.
- Two-way clustered standard errors by Store and Date: This is standard in store-week panel settings.

With store fixed effects and week fixed effects, the model is using changes within the same store over time, after removing anything that is common to all stores in the same week like national shopping patterns, holidays, or seasonal demand. So the temperature effect is identified from week-to-week temperature changes for a given store, not from differences between stores.

```
# 2. Load libraries
```

```
library(tidyverse)
library(lubridate)
library(fixest)
library(modelsummary)
library(scales)
library(knitr)
library(kableExtra)
```

```
options(scipen = 999)
set.seed(123)
```

```
# 3. Load the Data
```

```
walmart_raw <- readr::read_csv(
  "data/Walmart_Sales.csv",
  show_col_types = FALSE
)
```

```
# Show structure
str(walmart_raw)
```

```
spc_tbl_ [6,435 x 8] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ Store      : num [1:6435] 1 1 1 1 1 1 1 1 1 1 ...
 $ Date       : chr [1:6435] "5/2/2010" "12/2/2010" "19-02-2010" "26-02-2010" ...
 $ Weekly_Sales: num [1:6435] 1643691 1641957 1611968 1409728 1554807 ...
 $ Holiday_Flag: num [1:6435] 0 1 0 0 0 0 0 0 0 0 ...
 $ Temperature : num [1:6435] 42.3 38.5 39.9 46.6 46.5 ...
 $ Fuel_Price  : num [1:6435] 2.57 2.55 2.51 2.56 2.62 ...
 $ CPI         : num [1:6435] 211 211 211 211 211 ...
 $ Unemployment: num [1:6435] 8.11 8.11 8.11 8.11 8.11 ...
 - attr(*, "spec")=
 .. cols(
 ..   Store = col_double(),
 ..   Date = col_character(),
 ..   Weekly_Sales = col_double(),
 ..   Holiday_Flag = col_double(),
 ..   Temperature = col_double(),
 ..   Fuel_Price = col_double(),
 ..   CPI = col_double(),
 ..   Unemployment = col_double()
 .. )
 - attr(*, "problems")=<externalptr>
```

```
# Show first 6 rows
head(walmart_raw)
```

```
# A tibble: 6 x 8
  Store Date       Weekly_Sales Holiday_Flag Temperature Fuel_Price CPI
  <dbl> <chr>         <dbl>         <dbl>         <dbl>     <dbl> <dbl>
1     1  5/2/2010      1643691.         0          42.3       2.57  211.
2     1 12/2/2010      1641957.         1          38.5       2.55  211.
3     1 19-02-2010      1611968.         0          39.9       2.51  211.
4     1 26-02-2010      1409728.         0          46.6       2.56  211.
5     1  5/3/2010      1554807.         0          46.5       2.62  211.
6     1 12/3/2010      1439542.         0          57.8       2.67  211.
# i 1 more variable: Unemployment <dbl>
```

```
# Summary statistics
summary(walmart_raw)
```

Store	Date	Weekly_Sales	Holiday_Flag
Min. : 1	Length:6435	Min. : 209986	Min. :0.00000
1st Qu.:12	Class :character	1st Qu.: 553350	1st Qu.:0.00000
Median :23	Mode :character	Median : 960746	Median :0.00000
Mean :23		Mean :1046965	Mean :0.06993
3rd Qu.:34		3rd Qu.:1420159	3rd Qu.:0.00000
Max. :45		Max. :3818687	Max. :1.00000
Temperature	Fuel_Price	CPI	Unemployment
Min. : -2.06	Min. :2.472	Min. :126.1	Min. : 3.879
1st Qu.: 47.46	1st Qu.:2.933	1st Qu.:131.7	1st Qu.: 6.891
Median : 62.67	Median :3.445	Median :182.6	Median : 7.874
Mean : 60.66	Mean :3.359	Mean :171.6	Mean : 7.999
3rd Qu.: 74.94	3rd Qu.:3.735	3rd Qu.:212.7	3rd Qu.: 8.622
Max. :100.14	Max. :4.468	Max. :227.2	Max. :14.313

4. Data Cleaning and Preparation

```
# 4.1 Fixing the Date column
# The dataset contains dates written in different
# formats (some use dashes, some slashes).
# To avoid mistakes, I parse dates using multiple possible formats.
```

```
walmart <- walmart_raw %>%
  mutate(
    Date = parse_date_time(Date, orders = c("dmy", "mdy", "dmY", "mdY"))
  ) %>%
  filter(!is.na(Date)) %>%
  mutate(Date = as_date(Date)) %>%
  arrange(Store, Date)
```

```
# Quick date checks
range(walmart$Date)
```

```
[1] "2010-02-05" "2012-10-26"
```

```
# Check that observations are weekly within store
walmart %>%
  group_by(Store) %>%
  arrange(Date) %>%
  summarise(
    median_gap_days = median(as.numeric(Date - lag(Date))), na.rm = TRUE),
    pct_gap_7days = mean(as.numeric(Date - lag(Date)) == 7, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  summary()
```

	Store	median_gap_days	pct_gap_7days
Min.	: 1	Min. :7	Min. :1
1st Qu.:	12	1st Qu.:7	1st Qu.:1
Median	:23	Median :7	Median :1
Mean	:23	Mean :7	Mean :1
3rd Qu.:	34	3rd Qu.:7	3rd Qu.:1
Max.	:45	Max. :7	Max. :1

The median gap is 7 days and most gaps are 7 days, so the dates are correctly spaced.

```
# 4.2 Panel structure check
```

```
cat("Rows:", nrow(walmart), "\n")
```

```
Rows: 6435
```

```
cat("Stores:", n_distinct(walmart$Store), "\n")
```

```
Stores: 45
```

```
cat("Weeks:", n_distinct(walmart$Date), "\n")
```

```
Weeks: 143
```

```
walmart %>%
  group_by(Store) %>%
  summarise(n_weeks = n_distinct(Date), .groups = "drop") %>%
  summary()
```

	Store	n_weeks
Min.	: 1	Min. :143
1st Qu.:	12	1st Qu.:143
Median	:23	Median :143
Mean	:23	Mean :143
3rd Qu.:	34	3rd Qu.:143
Max.	:45	Max. :143

All stores appear in most weeks which shows a balanced panel.

```
# 4.3 Log sales
```

```
walmart <- walmart %>%
  mutate(Log_Sales = log(Weekly_Sales))
```

Weekly sales are large and skewed. Taking logs makes the distribution more stable and turns coefficients into approximate percent changes, which is easier to interpret.

```
# 4.4 Note about Holiday_Flag with Date fixed effects
```

```
walmart %>%
  group_by(Date) %>%
  summarise(n_holiday_values = n_distinct(Holiday_Flag), .groups = "drop") %>%
  count(n_holiday_values)
```

```
# A tibble: 1 x 2
```

	n_holiday_values	n
	<int>	<int>
1	1	143

In this dataset, Holiday_Flag is the same for all stores in the same week.

If we include Date fixed effects, Holiday_Flag is perfectly collinear (absorbed by the date dummies). We can quickly verify that.

Because every date has only one holiday value, I do not include Holiday_Flag separately in models that have Date FE.

```
# 5. Summary Statistics
```

```
walmart %>%
  summarise(
    mean_sales = mean(Weekly_Sales, na.rm = TRUE),
    sd_sales    = sd(Weekly_Sales, na.rm = TRUE),
    mean_temp   = mean(Temperature, na.rm = TRUE),
    sd_temp     = sd(Temperature, na.rm = TRUE),
    mean_fuel   = mean(Fuel_Price, na.rm = TRUE),
```

```

mean_cpi    = mean(CPI, na.rm = TRUE),
mean_unemp  = mean(Unemployment, na.rm = TRUE)
)

```

A tibble: 1 x 7

	mean_sales	sd_sales	mean_temp	sd_temp	mean_fuel	mean_cpi	mean_unemp
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1046965.	564367.	60.7	18.4	3.36	172.	8.00

6. Exploratory Graphs - Guiding the Model

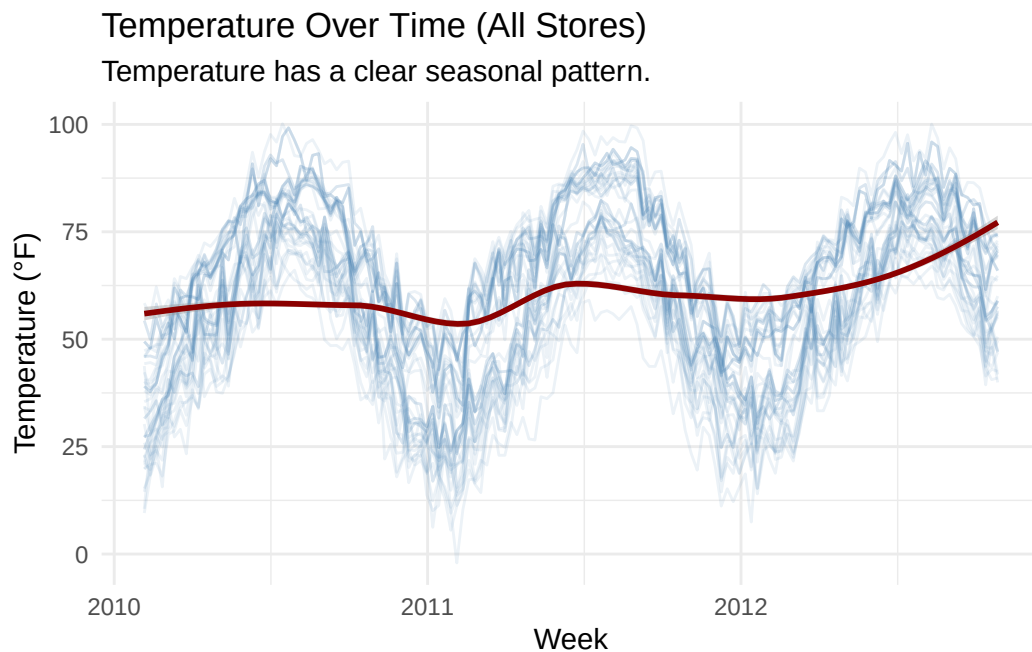
These graphs help us see why strong time controls are needed.

6.1 Temperature over time

```

ggplot(walmart, aes(x = Date, y = Temperature, group = Store)) +
  geom_line(alpha = 0.10, color = "steelblue") +
  geom_smooth(aes(group = 1), method = "loess", se = TRUE, color = "darkred") +
  labs(
    title = "Temperature Over Time (All Stores)",
    subtitle = "Temperature has a clear seasonal pattern.",
    x = "Week",
    y = "Temperature (°F)"
  ) +
  theme_minimal()

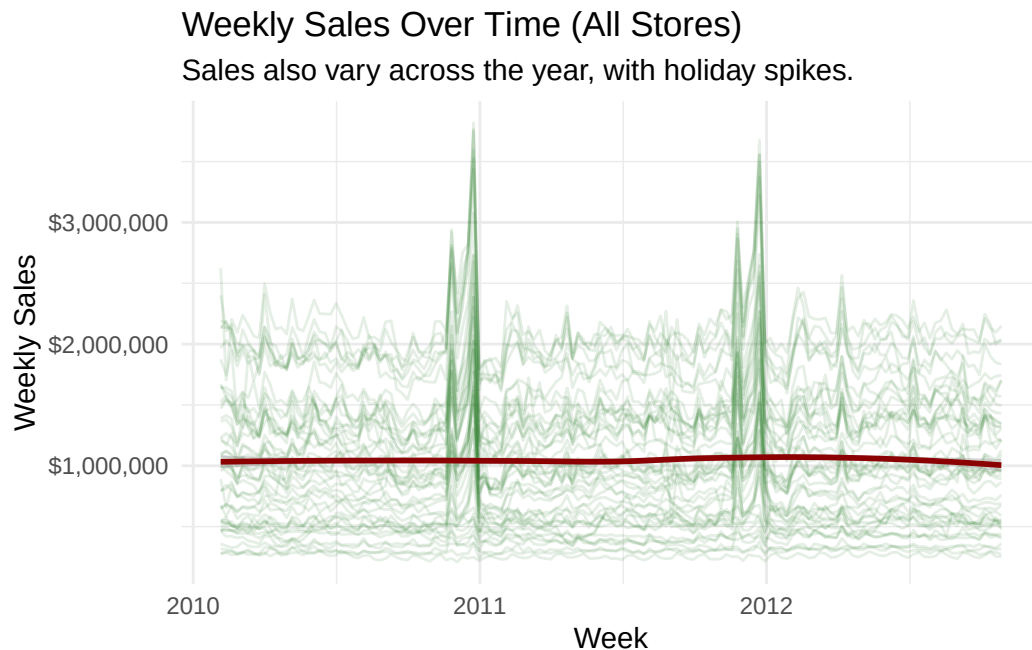
```



Temperature peaks in summer and lowers in winter, it shows a strong seasonal cycle.

```
# 6.2 Sales over time
```

```
ggplot(walmart, aes(x = Date, y = Weekly_Sales, group = Store)) +  
  geom_line(alpha = 0.10, color = "darkgreen") +  
  geom_smooth(aes(group = 1), method = "loess", se = TRUE, color = "darkred") +  
  labs(  
    title = "Weekly Sales Over Time (All Stores)",  
    subtitle = "Sales also vary across the year, with holiday spikes.",  
    x = "Week",  
    y = "Weekly Sales"  
  ) +  
  scale_y_continuous(labels = dollar) +  
  theme_minimal()
```



Sales also have seasonal patterns and big holiday bumps. If we want the pure temperature effect, we must control for these common time patterns, which is exactly what date fixed effects do.

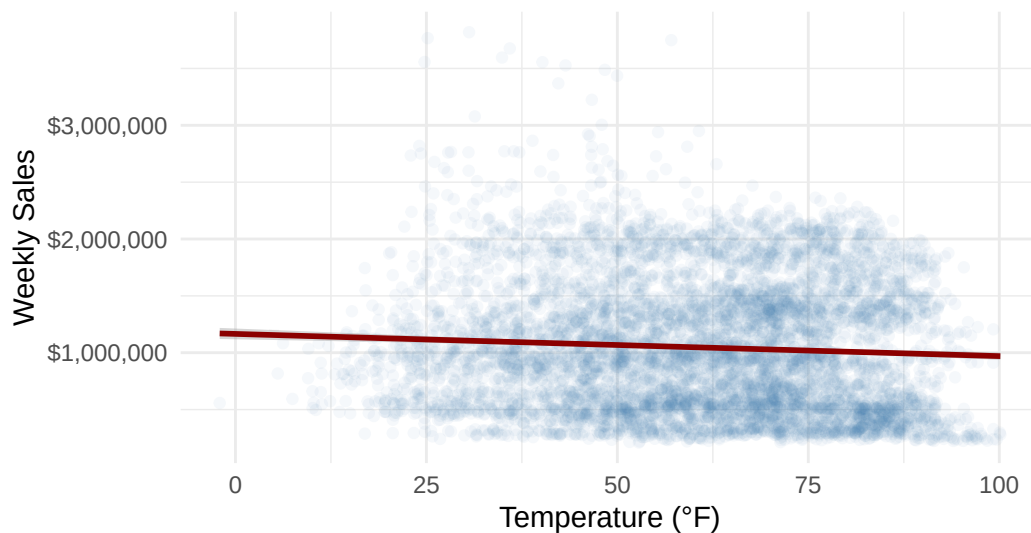
```
# 6.3 Raw relationship between sales and temperature
```

```
ggplot(walmart, aes(x = Temperature, y = Weekly_Sales)) +  
  geom_point(alpha = 0.05, color = "steelblue") +  
  geom_smooth(method = "lm", se = TRUE, color = "darkred") +  
  labs(  
    title = "Raw Sales vs. Temperature",  
    subtitle = "Without controls, sales look higher when it's warmer,  
    but that could be seasonality.",  
    x = "Temperature (°F)",  
    y = "Weekly Sales"  
  )
```

```
) +
scale_y_continuous(labels = dollar) +
theme_minimal()
```

Raw Sales vs. Temperature

Without controls, sales look higher when it's warmer, but that could be seasonality.



The raw plot suggests higher sales in warmer weeks. But we already know both variables move with seasons so this is likely spurious. The scatter also looks slightly curved, so I later test a quadratic specification to see whether the relationship departs from linearity once fixed effects are included.

7. Thinking About Causal Identification To answer “causally”, we need to imagine comparing two weeks in the same store that are identical in every way except temperature. Store fixed effects remove all time-invariant store differences. Date fixed effects remove all national-level shocks and seasonality.

What’s Left: The remaining variation comes from weeks where one store happened to be warmer or colder than the average for that week across all stores. That variation is plausibly more “as if random” after we condition on the controls.

But what about the other controls (Fuel_Price, CPI, Unemployment)? These variables could be:

Confounders: Things that affect both temperature and sales (e.g., economic conditions might be correlated with weather patterns).

Mediators: Things that temperature affects, which then affect sales (e.g., higher temperatures could increase fuel demand and raise fuel prices).

If they are mediators, controlling for them would bias the temperature coefficient (because we would be blocking part of the causal path). If they are confounders, we should control for them.

In practice, it’s hard to know for sure. I will include them in the main model because they capture some local economic variation that is not already absorbed by date fixed effects (date FE only).

removes national trends). Later, as a robustness check, I will run the model without them to see if the temperature estimate changes.

One detail is important here, because the model includes Date fixed effects, any purely national movement in fuel prices, CPI, or unemployment is already absorbed by the week dummies. These controls only matter through differences across stores within the same week, which helps account for local economic conditions that could be correlated with local weather shocks.

Causal story (Informal DAG):

Temperature → May affect sales directly (people might shop less when very hot or cold).

Temperature → May affect fuel prices (demand for energy) which could affect sales.

Temperature → May affect unemployment (e.g., seasonal jobs) which could affect sales.

Seasonality → Affects both temperature and sales. (Date FE removes this.)

Local economic conditions → Could affect sales and might be correlated with temperature e.g., a heat wave might coincide with a local economic boom. Fuel, CPI, Unemployment help control for this.

So the main model will include Fuel_Price, CPI, Unemployment, plus store and date fixed effects.

```
# 8. Main Fixed Effects Model

m_main <- feols(
  Log_Sales ~ Temperature + Fuel_Price + CPI + Unemployment | Store + Date,
  data = walmart,
  cluster = ~ Store + Date
)
```

```
# 9. Checking for Non-linearity

walmart <- walmart %>%
  mutate(Temp_sq = Temperature^2)

m_quad <- feols(
  Log_Sales ~ Temperature + Temp_sq + Fuel_Price + CPI +
    Unemployment | Store + Date, data = walmart,
  cluster = ~ Store + Date
)
```

To check whether the temperature effect is nonlinear, I added a quadratic term (Temperature²). In my results, the quadratic term is statistically significant, which suggests the relationship is not perfectly linear. That said, the quadratic coefficient is extremely small, so the curvature is mild in practice. We can say that sales change with temperature, but the shape of that relationship does not bend sharply across the temperature range in this dataset. So I treat the quadratic term as evidence of a small nonlinear pattern, but the overall temperature effect is still modest.

The quadratic coefficient appears as 0.000 in the table due to rounding however, it is small in magnitude but statistically significant.

```
# 10. Check for Lagged Effects

walmart <- walmart %>%
  group_by(Store) %>%
  arrange(Date) %>%
  mutate(
    Temp_lag1 = lag(Temperature, 1)
  ) %>%
  ungroup()

m_lag <- feols(
  Log_Sales ~ Temperature + Temp_lag1 + Fuel_Price + CPI +
  Unemployment | Store + Date, data = walmart,
  cluster = ~ Store + Date
)

summary(m_lag)
```

```
OLS estimation, Dep. Var.: Log_Sales
Observations: 6,390
Fixed-effects: Store: 45, Date: 142
Standard-errors: Clustered (Store & Date)
```

	Estimate	Std. Error	t value	Pr(> t)	
Temperature	0.001436	0.000587	2.448255	0.0184118	*
Temp_lag1	0.001570	0.000509	3.087563	0.0034878	**
Fuel_Price	-0.002694	0.028410	-0.094813	0.9248939	
CPI	0.000157	0.005448	0.028777	0.9771729	
Unemployment	-0.024630	0.012756	-1.930868	0.0599586	.

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.081909      Adj. R2: 0.98001
                Within R2: 0.029035
```

I also tested whether temperature affects sales with a delay by adding last week's temperature to the model. The lagged temperature coefficient is statistically significant at the 1% level. This suggests that temperature may have a small carryover effect into the following week. The magnitude of the coefficient is about 0.00157 which implies that a one-degree increase in temperature last week is associated with roughly a 0.157% increase in sales this week, holding store and week fixed effects constant.

Even though the effect is statistically significant, it remains economically small. Overall, the results suggest that temperature influences sales mainly in the current week, with a modest additional impact from the previous week.

```
# 11. Robustness Checks
# 11.1 Without economic controls
# If fuel price, CPI, and unemployment are mediators, we should see the
```

```
# temperature coefficient change when we drop them.
```

```
m_no_controls <- feols(  
  Log_Sales ~ Temperature | Store + Date,  
  data = walmart,  
  cluster = ~ Store + Date  
)
```

```
# 11.2 Adding store-specific time trends  
# Sometimes stores have their own trends  
# (e.g., a store growing faster than others).  
# I add store-specific linear time trends to see if that changes the result.
```

```
walmart <- walmart %>%  
  group_by(Store) %>%  
  mutate(week_id = as.numeric(Date - min(Date)) / 7) %>%  
  ungroup()  
  
m_trend <- feols(  
  Log_Sales ~ Temperature + Fuel_Price + CPI + Unemployment | Store[week_id] +  
  Date, data = walmart, cluster = ~ Store + Date  
)
```

```
# 12. Results Tables
```

```
models_1 <- list(  
  "Main" = m_main,  
  "No controls" = m_no_controls,  
  "Quadratic" = m_quad  
)  
  
models_2 <- list(  
  "With lag" = m_lag,  
  "Store trends" = m_trend  
)  
  
common_args <- list(  
  stars = c("*" = .1, "**" = .05, "***" = .01),  
  coef_map = c(  
    "Temperature" = "Temperature",  
    "Temp_sq" = "Temperature2",  
    "Temp_lag1" = "Temperature (1-week lag)"  
  ),  
  gof_map = c("nobs", "r.squared", "vcov.type"),  
  notes = paste(  
    "All models include Store and Date fixed effects, except where noted.",  
    "Standard errors clustered by Store and Date."
```

Table 1: Regression Results (1/2): Main, No Controls, Quadratic

	Main	No controls	Quadratic
Temperature	0.002*** (0.001)	0.002*** (0.001)	0.005*** (0.001)
Temperature ²			0.000*** (0.000)
Num.Obs.	6435	6435	6435
R2	0.980	0.980	0.981
Std.Errors	by: Store & Date	by: Store & Date	by: Store & Date

* p <0.1, ** p <0.05, *** p <0.01

All models include Store and Date fixed effects, except where noted. Standard errors clustered by Store and Date.

Table 2: Regression Results (2/2): With Lag and Store Trends

	With lag	Store trends
Temperature	0.001** (0.001)	0.002*** (0.001)
Temperature (1-week lag)	0.002*** (0.001)	
Num.Obs.	6390	6435
R2	0.981	0.985
Std.Errors	by: Store & Date	by: Store & Date

* p <0.1, ** p <0.05, *** p <0.01

All models include Store and Date fixed effects, except where noted. Standard errors clustered by Store and Date.

```
)
)

do.call(modelsummary, c(list(
  models_1,
  title = "Regression Results (1/2): Main, No Controls, Quadratic"
), common_args))
```

```
do.call(modelsummary, c(list(
  models_2,
  title = "Regression Results (2/2): With Lag and Store Trends"
), common_args))
```

```
# 13. Interpreting the Results
```

```
b_temp <- coef(m_main)["Temperature"]  
pct_change <- 100 * (exp(b_temp) - 1)
```

```
ci <- confint(m_main, parm = "Temperature")  
ci_pct <- 100 * (exp(ci) - 1)
```

```
cat("Temperature coefficient (log points): ", round(b_temp, 6), "\n", sep = "")
```

```
Temperature coefficient (log points): 0.00199
```

```
cat("Approx. percent change in sales per +1°F: ",  
    round(pct_change, 3), "%\n", sep = "")
```

```
Approx. percent change in sales per +1°F: 0.199%
```

```
cat("95% CI in percent terms: [", round(ci_pct[[1]], 3), "%, ",  
    round(ci_pct[[2]], 3), "%]\n", sep = "")
```

```
95% CI in percent terms: [0.076%, 0.323%]
```

In the log model, the temperature coefficient translates into about a 0.199% increase in weekly sales for a 1°F increase in temperature, holding the other controls and fixed effects constant. The 95% confidence interval is roughly 0.076% to 0.323%, so the estimate is positive and fairly precise statistically. Even so, the effect size is small in business terms: a one-degree change in temperature is not expected to meaningfully change sales on its own.

Now we look at the other models.

No controls gives a slightly larger coefficient (about 0.0020). This suggests that adding Fuel_Price, CPI, and Unemployment explains a small part of the variation, but the temperature effect remains very similar.

Quadratic: The squared term is extremely small, and it does not change the main conclusion. Any non linearity appears mild in practical terms.

With lag: The lagged temperature coefficient is statistically significant, which suggests a small carryover effect into the next week. However, the magnitude is still modest, so the economic impact remains small.

Store trends: The coefficient barely changes (around 0.0016), which suggests store-specific trends are not driving the result.

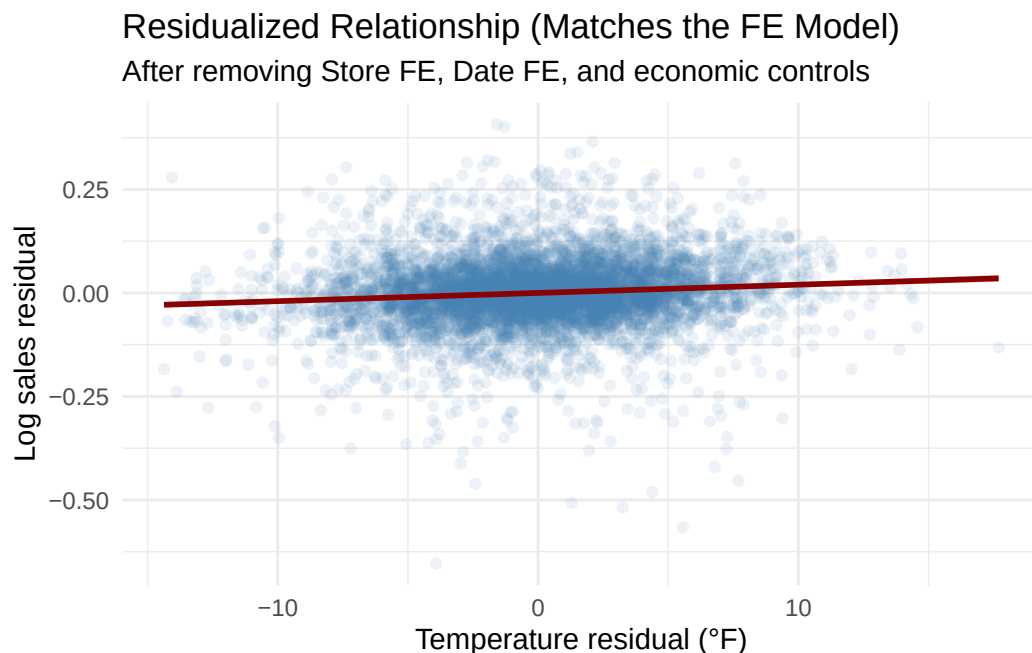
Overall, the temperature estimate is stable across specifications, and the main conclusion stays the same: the effect is positive but small.

```
# 14. Visual Check of the Residual Relationship
# This plot shows the relationship that the fixed-effects model actually uses,
# after removing store effects, date effects, and controls from both variables.

resid_sales <- resid(feols(Log_Sales ~ Fuel_Price + CPI +
                          Unemployment | Store + Date, data = walmart))
resid_temp  <- resid(feols(Temperature ~ Fuel_Price + CPI +
                          Unemployment | Store + Date, data = walmart))

plot_df <- tibble(resid_sales = resid_sales, resid_temp = resid_temp)

ggplot(plot_df, aes(x = resid_temp, y = resid_sales)) +
  geom_point(alpha = 0.10, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "darkred") +
  labs(
    title = "Residualized Relationship (Matches the FE Model)",
    subtitle = "After removing Store FE, Date FE, and economic controls",
    x = "Temperature residual (°F)",
    y = "Log sales residual"
  ) +
  theme_minimal()
```



The line is slightly upward but very shallow, consistent with a small positive effect.

15. Answer to the Research Question Do higher temperatures causally affect Walmart weekly sales?

Based on the fixed-effects models:

The estimated effect is positive but very small, about a 0.20% increase in sales per 1°F (roughly 0.199% in the main model).

This result is robust to dropping controls, adding a quadratic term, including a lag and adding store-specific trends.

The effect is economically small compared to normal seasonal swings or holiday weeks.

So while temperature does have a statistically detectable effect, it is not large enough to be a major concern for Walmart's overall sales forecasting.

16. Limitations Even with store and date fixed effects, we cannot completely rule out local, time-varying confounders. For example, a local festival or a severe storm could affect both temperature perceptions and shopping behavior, and those events might not be fully captured by date fixed effects, because they hit only one store. That kind of bias would be small and unlikely to overturn the conclusion, but it's worth mentioning.

Also, the data covers only 45 stores over about 2.5 years, results might differ in other regions or longer periods.

17. Executive Summary Sales and temperature both follow seasonal patterns, so a simple comparison is misleading. After using store and week fixed effects to control for those patterns, I find that a 1°F warmer week is associated with only about a 0.20% increase in sales (about 0.199% in the main model). That's a tiny effect, much smaller than the usual holiday bumps. Walmart probably does not need to worry about gradual temperature changes when planning weekly sales.